TAIWAN OASIS

Http://www.taiwanoasis.com.tw

Technical Data

56.8mm (2.3") 7 Segment Display

TOS-23101X Single Digit

TOS-23102X Single Digit

TOD-23201X Dual Digit

Features

*(2.3")56.8mm DIGIT HEIGHT
* CONTINUOUS UNIFORM SEGMENTS
* LOW POWER REQUIREMENT
* EXCELLENT CHARACTER APPEARANCE
* HIGH BRIGHTNESS & HIGH CONTRAST
* WIDE VIEWING ANGLE
* SOLID STATE RELIABILITY
* RoHS COMPLIANT
* Pb FREE
* GRAY/BLACK FACE AVAILABLE

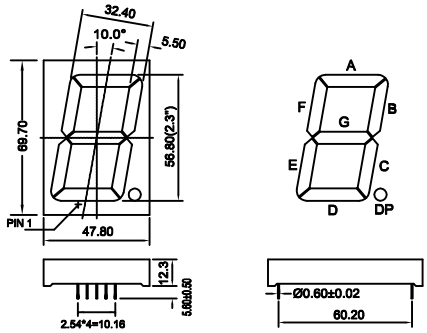


Absolute Maximum Rating

Series	Emitting color	Pd (mW)	If (mA)	IFP (mA)	VR (V)	Top (℃)	Tst (℃)	Tsol*2 (℃)	VF			λp	IV	
									IF (mA)	typ (V)	max (V)	typ (nm)	IF (mA)	typ (ucd)
2.3"	H	60	25	80	5	-25 ~ +85	-30 ~ +85	260/3Sec	20	1.8	2.0	660	10	9185
	S	60	25					260/3Sec	20	1.8	2.0	660	10	15524
	K	60	25					260/3Sec	20	1.8	2.0	648	10	9185
	E	60	25					260/3Sec	20	2.1	2.4	630	10	7066
	Y	60	25					260/3Sec	20	2.1	2.4	585	10	4113
	G	70	25					260/3Sec	20	2.2	2.5	570	10	7066

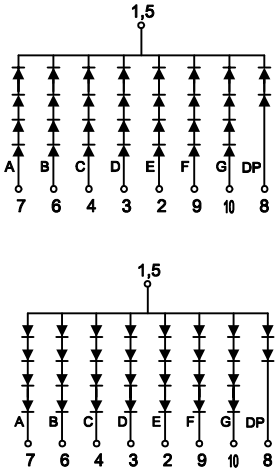
*1.Condition for IFP is pulse of 1 / 10 duty and 0.1 msec width.
*2. Pb Free Process.
*3. In the absence of confirmation by device specification sheets,OASIS takes no responsibility for any defects that may occur in equipment using any OASIS devices shown in catalogs, data books,etc. Contact OASIS in order to obtain the latest device specification sheets before using any OASIS devices.
*4. Specifictions are subject to change without notice for improvement.

PACKAGE DIMENSIONS

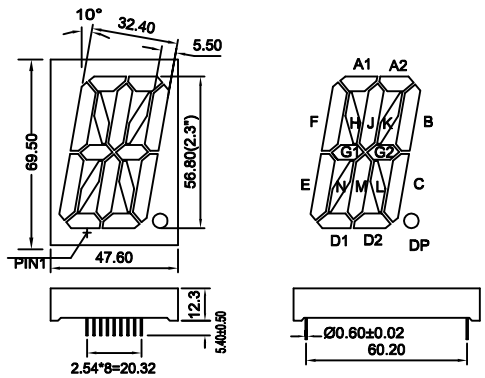


NORMAL TOLERANCE:±0.25 ANGLE:±1°

INTERNAL CIRCUIT DIAGRAM

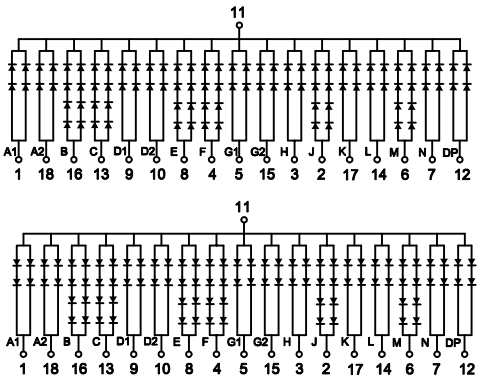


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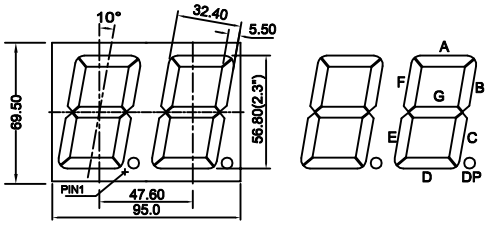


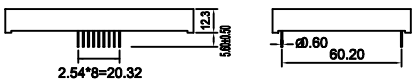
NORMAL TOLERANCE:±0.25 ANGLE:±1°

INTERNAL CIRCUIT DIAGRAM



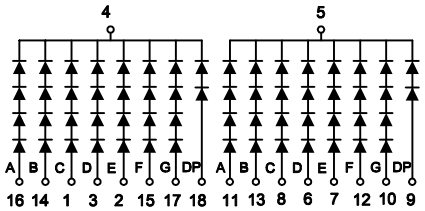
PACKAGE DIMENSIONS

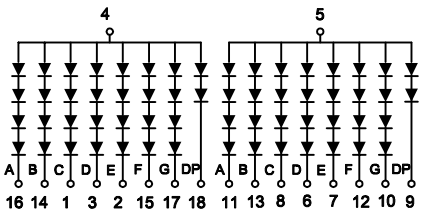




NORMAL TOLERANCE:±0.25 ANGLE:±1°

INTERNAL CIRCUIT DIAGRAM





Chip Material

H(Hi-red): GaAlAs/GaAs

S(Super-red): GaAlAs/GaAs

K(Hi-Orange): GaAlAs/GaAs

ABSOLUTE MAXIMUN RATING :(TA=25°C)

SYMBOL	PARAMETER	VALUE			UNITS
		H	S	K	
Pd	Power Dissipation Per Segment	60			mW
IFP	Peak Forward Current Per Segment (1/10 Duty Cycle, 0.1ms Pulse Width)	80			mA
IFM	Continuous Forward Current Per Segment	25			mA
TOP	Reverse Voltage Per Segment	5			V
TST	Operating Temperature Range	-25 to +85			°C
VR	Storage Temperature Range	-30 to +85			°C
Lead Solder Temperature(1/16 Inch Below Seating Plane)		260°C for 3 sec.			

ELECTRICAL/OPTICAL CHARACTERISTICS AT TA=25°C

SYMBOL	PARAMETER	TEST CONDITION		MIN.	TYP.	MAX.	UNITS
Iv	Luminous Intensity Per Segment	IF=10mA	H		9185		µcd
			S		15524		µcd
			K		9185		µcd
λp	Peak Emission Wavelength	IF=20mA	H		660		nm
			S		660		nm
			K		648		nm
λd	Domiant Emission Wavelength	IF=20mA	H		643		nm
			S		640		nm
			K		635		nm
△λ	Spectral Line Half-Width	IF=20mA			20		nm
VF	Forward Voltage Per Segment	IF=20mA		1.7	1.8	2.0	V
IR	Reverse Current Per Segment	VR=5V			10		µA
Iv-m	Luminous Intensity Matching Rate	IF=20mA			1.5:1		

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*Specifictions are subject to change without notice for improvement.

Chip Material

G(Green): GaP

R(Red): GaP

ABSOLUTE MAXIMUN RATING :(TA=25°C)

SYMBOL	PARAMETER	VALUE		UNITS
		G	R	
Pd	Power Dissipation Per Segment	50	50	mW
I_{FP}	Peak Forward Current Per Segment (1/10 Duty Cycle, 0.1ms Pulse Width)	80	80	mA
I_{FM}	Continuous Forward Current Per Segment	25	15	mA
T_{OP}	Reverse Voltage Per Segment	5		V
T_{ST}	Operating Temperature Range	-25 to +85		°C
V_R	Storage Temperature Range	-30 to +85		°C
Lead Solder Temperature(1/16 Inch Below Seating Plane)		260°C for 3 sec.		

ELECTRICAL/OPTICAL CHARACTERISTICS AT TA=25°C

SYMBOL	PARAMETER	TEST CONDITION		MIN.	TYP.	MAX.	UNITS
Iv	Luminous Intensity Per Segment	I_F=10mA	G		7066		µcd
			R		780		µcd
λ_p	Peak Emission Wavelength	I_F=20mA	G		570		nm
			R		700		nm
λ_d	Domiant Emission Wavelength	I_F=20mA	G		572		nm
			R		655		nm
$\triangle\lambda$	Spectral Line Half-Width	I_F=20mA	G		30		nm
			R		90		nm
V_F	Forward Voltage Per Segment	I_F=20mA		1.9	2.2	2.5	V
I_R	Reverse Current Per Segment	V_R=5V			10		µA
Iv-m	Luminous Intensity Matching Rate	I_F=20mA			1.5:1		

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Chip Material

A(Amber): GaAsP/GaP
E(Orange): GaAsP/GaP
Y(Yellow): GaAsP/GaP

ABSOLUTE MAXIMUN RATING :(TA=25°C)

SYMBOL	PARAMETER	VALUE			UNITS
		A	E	Y	
P_d	Power Dissipation Per Segment	65			mW
I_{FP}	Peak Forward Current Per Segment (1/10 Duty Cycle, 0.1ms Pulse Width)	80			mA
I_{FM}	Continuous Forward Current Per Segment	25			mA
T_{OP}	Reverse Voltage Per Segment	5			V
T_{ST}	Operating Temperature Range	-25 to +85			°C
V_R	Storage Temperature Range	-30 to +85			°C
Lead Solder Temperature(1/16 Inch Below Seating Plane)		260°C for 3 sec.			

ELECTRICAL/OPTICAL CHARACTERISTICS AT TA=25°C

SYMBOL	PARAMETER	TEST CONDITION		MIN.	TYP.	MAX.	UNITS
Iv	Luminous Intensity Per Segment	I_F=10mA	A		2143		μcd
			E		7066		μcd
			Y		4113		μcd
λ_P	Peak Emission Wavelength	I_F=20mA	A		610		nm
			E		630		nm
			Y		585		nm
λ_d	Domiant Emission Wavelength	I_F=20mA	A		608		nm
			E		621		nm
			Y		586		nm
$\triangle\lambda$	Spectral Line Half-Width	I_F=20mA			35		nm
V_F	Forward Voltage Per Segment	I_F=20mA		1.8	2.1	2.4	V
I_R	Reverse Current Per Segment	V_R=5V			10		μA
Iv-m	Luminous Intensity Matching Rate	I_F=20mA			1.5:1		

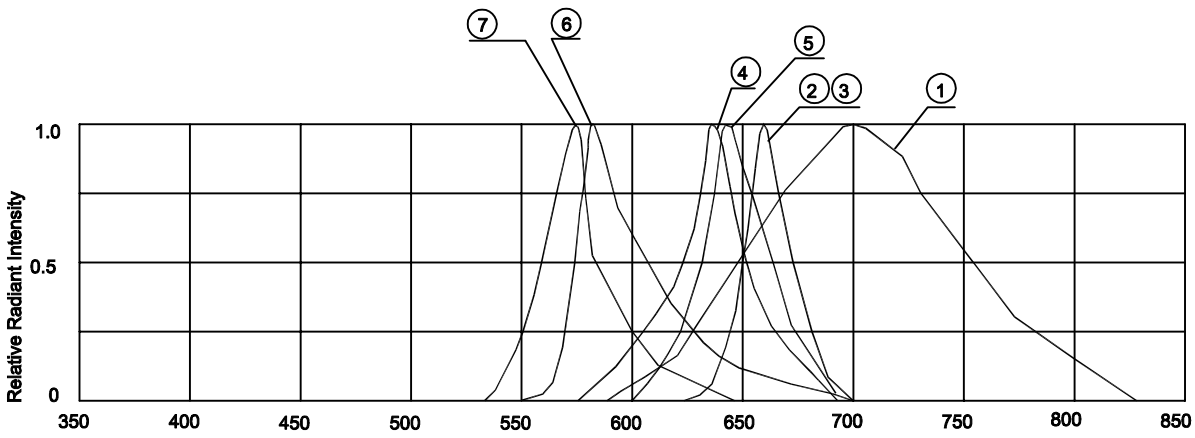
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TYPICAL ELECTRICAL/OPTICAL CHARACTERISTIC CURVES

Remark:

Dice code	①R	②H	③S	④K	⑤E	⑥Y	⑦G
Lighting color	Red	Hi-Red	Super-Red	Hi-Orange	Orange	Yellow	Green



TYPICAL ELECTRICAL/OPTICAL CHARACTERISTIC CURVES

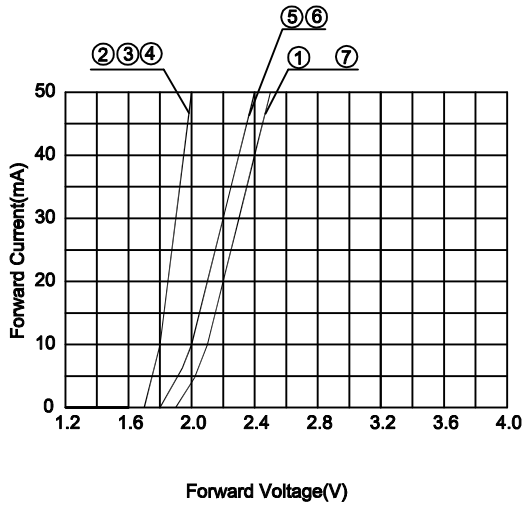


Fig2. FORWARD CURRENT VS.
FORWARD VOLTAGE

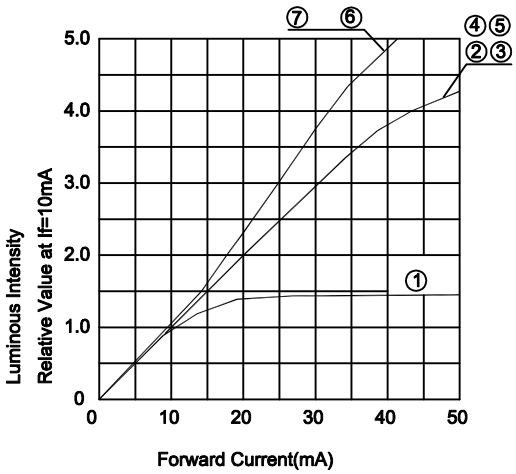
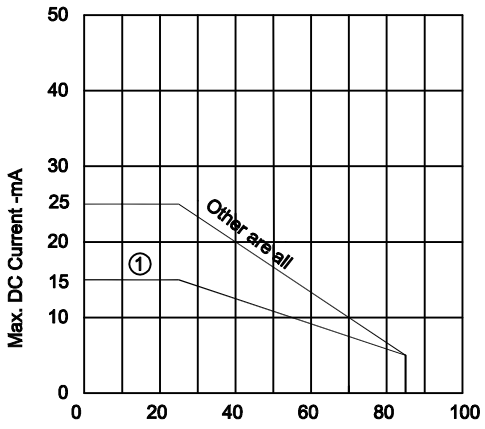


Fig3. RELATIVE LUMINOUS INTENSITY
VS. FORWARD CURRENT

Remark:

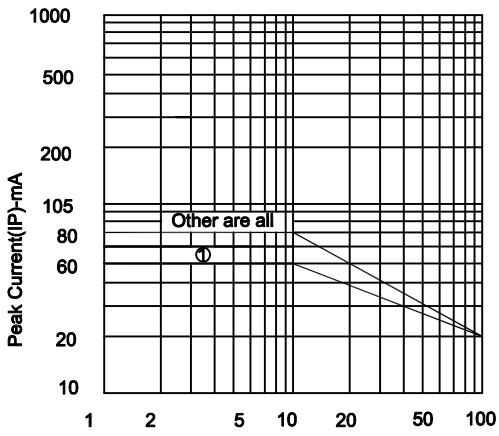
Dice code	①R	②H	③S	④K	⑤E	⑥Y	⑦G
Lighting color	Red	Hi-Red	Super-Red	Hi-Orange	Orange	Yellow	Green

TYPICAL ELECTRICAL/OPTICAL CHARACTERISTIC CURVES



Ambient Temperature TA (°C)

Fig4 MAX. ALLOWABLE DC CURRENT VS.AMBIENT TEMPERATURE.



Duty Cycle %

Fig5. MAX. PEAK CURRENT VS. DUTY CYCLE % (REFRESH RATE 1KHz)

Remark:

Dice code	①R	②H	③S	④K	⑤E	⑥Y	⑦G
Lighting color	Red	Hi-Red	Super-Red	Hi-Orange	Orange	Yellow	Green